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Roll#SU92-BSSEM-F25-041

ASSIGNMENT #03(PART 1)

PROGRAM#01

```
#include<iostream>

#include<iomanip>

using namespace std;


double calculateAverage(int testScores[],int size){

    int totalSum=0;


    for(int i=0;i<size;i++){

        totalSum+=testScores[i];

    }

    return (double)totalSum/size;

}





int main(){

    int numberofStudents;

    cout<<"Enter Number of Students"<<endl;
```

```
        cin>>numberofStudents;


        int*studentScores=new int[numberofStudents];


        cout<<numberofStudents<<"Enter Marks of each student"<<endl;


        for(int i=0;i<numberofStudents;i++)
        {
        cin>>studentScores[i];
}


        double classAverage=calculateAverage(studentScores,numberofStudents);
        cout<<fixed<<setprecision(2);
        cout<<"average score of a class is "<<classAverage<<endl;
        return 0;
}
```

PROGRAM#02

```
#include<iostream>
using namespace std;


int calculateLowest(int testScores[],int size){
        int lowest=testScores[0];
```

```
        for(int i=1;i<size;i++){
                if(testScores[i]<lowest){
                        lowest=testScores[i];
                }




        }
        return lowest;
}
int calculateHighest(int testScores[],int size){
        int highest=testScores[0];
        for(int i=1;i<size;i++){
                if(testScores[i]>highest){
                        highest=testScores[i];
                }

        }
        return highest;
}



int main(){
        int numberofStudents;
```

```
        cout<<"Enter numberofStudents"<<endl;

        cin>>numberofStudents;


        int scores[numberofStudents];

        cout<<"Enter scores";

        for(int i=0;i<numberofStudents;i++){

                cin>>scores[i];




        }


        cout<<"Lowest Score:"<<calculateLowest(scores,numberofStudents)<<endl;

                cout<<"Highest
Score:"<<calculateHighest(scores,numberofStudents)<<endl;

return 0;

}
```

PROGRAM#03

```
#include<iostream>

using namespace std;

int calculatePower(int base,int exponent){

        int result=1;

        for(int i=0;i<exponent;i++){

                result =result*base;

        }
```

```
        return result;
}
int main(){
        int base,exponent;
        for(; ;){

        cout<<"Enter Base and Exponent"<<endl;
        cin>>base>>exponent;
        int ans=calculatePower(base,exponent);
        cout<<base<<"raised to the power"<<exponent<<"is"<<ans;
}
}
```

PROGRAM#04

```
#include<iostream>
using namespace std;
int calculateVoltage(int current,int resistance){
        cout<<"Enter Current and Resistance"<<endl;
        cin>>current>>resistance;
        cout<<"The Total Voltage is"<<current*resistance<<endl;
}
int main(){
        int current,resistance;
```

```
        calculateVoltage(current,resistance);


}
```

PROGRAM#05

```
#include<iostream>
using namespace std;
bool initialFinder(char teamInitials[],int size,char targetInitial){
        for(int i=0;i<size;i++){
                if(teamInitials[i]==targetInitial){
                        return true;
                }
        }
                        return false;


        }


        int main(){


                char playingEleven[11]={'a','b','d','f','g','t','u','i','p','q','c'};
                char targetInitial;
                cout<<"Enter Players initial"<<endl;
                cin>>targetInitial;
                if (initialFinder(playingEleven,11,targetInitial)){
```

```
                cout<<"The Player is in the team"<<endl;

        }else{
                cout<<"Player not found"<<endl;
        }
}
```

PROGRAM#06

```
#include<iostream>
using namespace std;
int calculateLength(char word[]){
    int count=0;
    while(word[count]!='\0'){
        count++;
    }
    return count;
}
int main(){
    char word[100];
    cout<<"Enter Word"<<endl;
    cin>>word;
    int length=calculateLength(word);
    cout<<"The length of a word is "<<length<<endl;
}
```